

Using Boole's inequality, we obtain

$$\begin{aligned}
 & \frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda,2}^b \left(\mu^{(b,2)} | \mathbb{X}_m, \mathbb{Y}_n \right) - \widehat{q}_{1-u^*w_\lambda}^{\lambda, B_1} (\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \\
 & \leq \sum_{\lambda \in \Lambda} \mathbb{P}_{p \times p \times r} \left(\widehat{\text{MMD}}_\lambda^2 (\mathbb{X}_m, \mathbb{Y}_n) > \widehat{q}_{1-u^*w_\lambda}^{\lambda, B_1} (\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) + \sqrt{\frac{1}{2B_2} \ln \left(\frac{2}{\beta} \right)} \\
 & \leq \sum_{\lambda \in \Lambda} u^* w_\lambda + \sqrt{\frac{1}{2B_2} \ln \left(\frac{2}{\beta} \right)} \\
 & \leq u^* + \sqrt{\frac{1}{2B_2} \ln \left(\frac{2}{\beta} \right)} \\
 & = \frac{3\alpha}{4} + \sqrt{\frac{1}{2B_2} \ln \left(\frac{2}{\beta} \right)}
 \end{aligned}$$

for $u^* := \frac{3\alpha}{4}$, where we have used Proposition 1 and the fact that $\sum_{\lambda \in \Lambda} w_\lambda \leq 1$. Now, for $B_2 \geq \frac{8}{\alpha^2} \ln \left(\frac{2}{\beta} \right)$, we get

$$\frac{3\alpha}{4} + \sqrt{\frac{1}{2B_2} \ln \left(\frac{2}{\beta} \right)} \leq \alpha$$

and so, we obtain

$$\frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda,2}^b \left(\mu^{(b,2)} | \mathbb{X}_m, \mathbb{Y}_n \right) - \widehat{q}_{1-u^*w_\lambda}^{\lambda, B_1} (\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \leq \alpha.$$

Recall that $u_\alpha^{B_2}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1})$ is defined as

$$\sup \left\{ u \in \left(0, \min_{\lambda \in \Lambda} w_\lambda^{-1} \right) : \frac{1}{B_2} \sum_{b=1}^{B_2} \mathbb{1} \left(\max_{\lambda \in \Lambda} \left(\widehat{M}_{\lambda,2}^b \left(\mu^{(b,2)} | \mathbb{X}_m, \mathbb{Y}_n \right) - \widehat{q}_{1-uw_\lambda}^{\lambda, B_1} (\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) > 0 \right) \leq \alpha \right\},$$

we deduce that

$$u_\alpha^{B_2}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) \geq u^* = \frac{3\alpha}{4}$$

for $B_2 \geq \frac{8}{\alpha^2} \ln \left(\frac{2}{\beta} \right)$ when the event $\mathcal{A}$ holds.

Under the event $\mathcal{A}$, after performing B_3 steps of the bisection method, we have

$$\begin{aligned}
 \widehat{u}_\alpha^{B_2:3}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) & \geq u_\alpha^{B_2}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}) - \frac{\min_{\lambda \in \Lambda} w_\lambda^{-1}}{2B_3} \\
 & \geq \frac{3\alpha}{4} - \frac{\min_{\lambda \in \Lambda} w_\lambda^{-1}}{2B_3} \\
 & \geq \frac{\alpha}{2}
 \end{aligned}$$

for $B_3 \geq \log_2 \left(\frac{4}{\alpha} \min_{\lambda \in \Lambda} w_\lambda^{-1} \right)$.

We are interested in upper bounding the probability of type II error $\mathbb{P}_{p \times q \times r \times r}(\mathcal{B})$ for the event $\mathcal{B} := \left\{ \Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = 0 \right\}$. We have

$$\begin{aligned} \mathbb{P}_{p \times q \times r \times r}(\mathcal{B}) &= \mathbb{P}_{p \times q \times r \times r}(\mathcal{B} | \mathcal{A}) \mathbb{P}_{p \times q \times r \times r}(\mathcal{A}) + \mathbb{P}_{p \times q \times r \times r}(\mathcal{B} | \mathcal{A}^c) \mathbb{P}_{p \times q \times r \times r}(\mathcal{A}^c) \\ &\leq \mathbb{P}_{p \times q \times r \times r}(\mathcal{B} | \mathcal{A}) + \frac{\beta}{2} \end{aligned}$$

where

$$\begin{aligned} &\mathbb{P}_{p \times q \times r \times r}(\mathcal{B} | \mathcal{A}) \\ &= \mathbb{P}_{p \times q \times r \times r} \left(\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = 0 \mid \mathcal{A} \right) \\ &= \mathbb{P}_{p \times q \times r \times r} \left(\bigcap_{\lambda \in \Lambda} \left\{ \widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\hat{u}_\alpha^{B_{2:3}}}^{\lambda, B_1}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1})_{w_\lambda}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right\} \mid \mathcal{A} \right) \\ &\leq \min_{\lambda \in \Lambda} \mathbb{P}_{p \times q \times r \times r} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\hat{u}_\alpha^{B_{2:3}}}^{\lambda, B_1}(\mathbb{Z}_{B_2} | \mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1})_{w_\lambda}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \mid \mathcal{A} \right) \\ &\leq \min_{\lambda \in \Lambda} \mathbb{P}_{p \times q \times r} \left(\widehat{\text{MMD}}_\lambda^2(\mathbb{X}_m, \mathbb{Y}_n) \leq \hat{q}_{1-\alpha w_\lambda/2}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) \right) \\ &= \min_{\lambda \in \Lambda} \mathbb{P}_{p \times q \times r} \left(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) = 0 \right), \end{aligned}$$

we deduce that

$$\mathbb{P}_{p \times q \times r \times r}(\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = 0) \leq \frac{\beta}{2} + \min_{\lambda \in \Lambda} \mathbb{P}_{p \times q \times r} \left(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) = 0 \right).$$

In order to upper bound $\mathbb{P}_{p \times q \times r \times r}(\Delta_\alpha^{\Lambda^w, B_{1:3}}(\mathbb{X}_m, \mathbb{Y}_n, \mathbb{Z}_{B_1}, \mathbb{Z}_{B_2}) = 0)$ by β it is sufficient to upper bound $\min_{\lambda \in \Lambda} \mathbb{P}_{p \times q \times r}(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}(\mathbb{Z}_{B_1} | \mathbb{X}_m, \mathbb{Y}_n) = 0)$ by $\frac{\beta}{2}$. By definition of uniform separation rates, it follows that

$$\rho(\Delta_\alpha^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M)^2 \leq 4 \min_{\lambda \in \Lambda} \rho(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}, \mathcal{S}_d^s(R), \beta, M)^2.$$

For each $\lambda \in \Lambda$, since $(1 - \alpha w_\lambda/2)\alpha w_\lambda/2 \leq (1 - \alpha)\alpha$ as $\alpha \in (0, e^{-1})$ and $w_\lambda \leq 1$, we have

$$\begin{aligned} B_1 &\geq \left(\max_{\lambda \in \Lambda} w_\lambda^{-2} \right) \frac{12}{\alpha^2} \left(\log \left(\frac{8}{\beta} \right) + \alpha(1 - \alpha) \right) \\ &\geq \frac{3}{(\alpha w_\lambda/2)^2} \left(\log \left(\frac{8}{\beta} \right) + \frac{\alpha w_\lambda}{2} \left(1 - \frac{\alpha w_\lambda}{2} \right) \right), \end{aligned}$$

so we can apply Theorem 6 to the tests $\left(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}\right)_{\lambda \in \Lambda}$ to obtain

$$\begin{aligned} \rho\left(\Delta_{\alpha}^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M\right)^2 &\leq 4 \min_{\lambda \in \Lambda} \rho\left(\Delta_{\alpha w_\lambda/2}^{\lambda, B_1}, \mathcal{S}_d^s(R), \beta, M\right)^2 \\ &\leq 4C_4(M, d, s, R, \beta) \min_{\lambda \in \Lambda} \left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln\left(\frac{2}{\alpha w_\lambda}\right)}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\ &\leq C_6(M, d, s, R, \beta) \min_{\lambda \in \Lambda} \left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln\left(\frac{1}{\alpha w_\lambda}\right)}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \end{aligned}$$

for $C_6(M, d, s, R, \beta) := 8C_4(M, d, s, R, \beta)$ where $C_4(M, d, s, R, \beta)$ is the constant from Theorem 6, and where we used the fact that

$$\ln\left(\frac{2}{\alpha w_\lambda}\right) = \ln(2) + \ln\left(\frac{1}{\alpha w_\lambda}\right) \leq (\ln(2) + 1) \ln\left(\frac{1}{\alpha w_\lambda}\right) \leq 2 \ln\left(\frac{1}{\alpha w_\lambda}\right)$$

as $\ln\left(\frac{1}{\alpha w_\lambda}\right) \geq \ln\left(\frac{1}{\alpha}\right) > 1$ since $\alpha \in (0, e^{-1})$.

E.10 Proof of Corollary 10

First, note that we indeed have

$$\sum_{\lambda \in \Lambda} w_\lambda < \frac{6}{\pi^2} \sum_{\ell=1}^{\infty} \frac{1}{\ell^2} = 1$$

and also that for all $\lambda = (2^{-\ell}, \dots, 2^{-\ell}) \in \Lambda$ we have $\lambda_1 \cdots \lambda_d = 2^{-d\ell} < 1$ as $\ell, d \in \mathbb{N} \setminus \{0\}$. Let $\lambda^* = (2^{-\ell^*}, \dots, 2^{-\ell^*}) \in \Lambda$ where

$$\ell^* := \left\lceil \frac{2}{4s+d} \log_2 \left(\frac{m+n}{\ln(\ln(m+n))} \right) \right\rceil \leq \left\lceil \frac{2}{d} \log_2 \left(\frac{m+n}{\ln(\ln(m+n))} \right) \right\rceil.$$

Since $\min_{\lambda \in \Lambda} w_\lambda^{-1} = \frac{6}{\pi^2}$, we have $B_3 \geq \log_2 \left(\frac{4}{\alpha} \min_{\lambda \in \Lambda} w_\lambda^{-1} \right)$, so we can apply Theorem 9 to get

$$\begin{aligned} \rho\left(\Delta_{\alpha}^{\Lambda^w, B_{1:3}}, \mathcal{S}_d^s(R), \beta, M\right)^2 &\leq C_6(M, d, s, R, \beta) \min_{\lambda \in \Lambda} \left(\sum_{i=1}^d \lambda_i^{2s} + \frac{\ln\left(\frac{1}{\alpha}\right) + \ln\left(\frac{1}{w_\lambda}\right)}{(m+n) \sqrt{\lambda_1 \cdots \lambda_d}} \right) \\ &\leq C_6(M, d, s, R, \beta) \left(\sum_{i=1}^d (\lambda_i^*)^{2s} + \frac{\ln\left(\frac{1}{\alpha}\right) + \ln\left(\frac{1}{w_{\lambda^*}}\right)}{(m+n) \sqrt{\lambda_1^* \cdots \lambda_d^*}} \right). \end{aligned}$$

Note that $\ell^* \leq \frac{2}{4s+d} \log_2 \left(\frac{m+n}{\ln(\ln(m+n))} \right) + 1$ which gives $\lambda_i^* = 2^{-\ell^*} \geq 2^{-1} \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-2/(4s+d)}$

for $i = 1, \dots, d$. We get $\sqrt{\lambda_1^* \cdots \lambda_d^*} \geq 2^{-\frac{d}{2}} \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{-d/(4s+d)}$ and so

$$\frac{1}{\sqrt{\lambda_1^* \cdots \lambda_d^*}} \leq 2^{\frac{d}{2}} \left(\frac{m+n}{\ln(\ln(m+n))} \right)^{d/(4s+d)}.$$